Lecture 4. Trade and Resources: The Heckscher-Ohlin Model

Lectures on International Trade

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Recall from Lecture 2 and 3

Gains and losses from trade:

- Ricardian model
  - One factor: labor.
  - All labor gains from trade.
- Specific-factors model
  - Mobile factor and specific factors.
  - Ambiguous for mobile factor, some specific factor gain, some lose.
  - A short-run model
- A long-run model: the Heckscher-Ohlin model

Outline

- The Heckscher-Ohlin Model
  - Assumptions
  - No-trade equilibrium
  - Free-trade equilibrium
  - Heckscher-Ohlin Theorem
- Testing the Model
  - The Leontief's paradox
- Effects of Trade on Factor Prices
  - Stolper-Samuelson Theorem

Motivating Facts

- Different countries have different resource endowments.
  - Natural resources, climate, geography
  - Labor, capital, land
- Factors of production can move between industries in the long-run.

The Heckscher-Ohlin Model

Assumptions of the Heckscher-Ohlin Model

- Assumption 1: Two factors of production, labor and capital, can move freely between the industries.

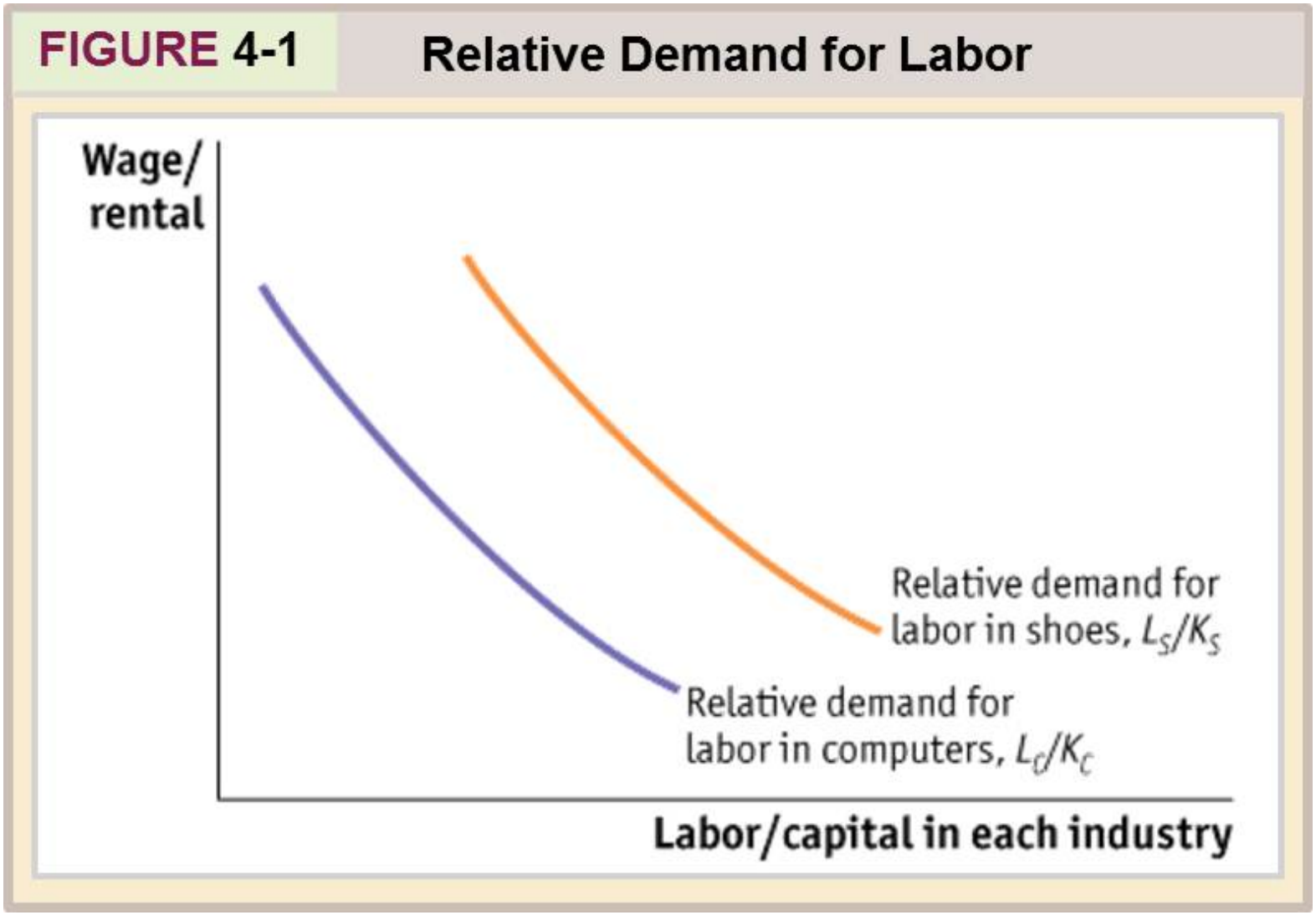
The Heckscher-Ohlin Model

Assumptions of the Heckscher-Ohlin Model

- Assumption 1: Two factors of production, labor and capital, can move freely between the industries.
- Assumption 2: Shoe production is labor-intensive; that is, it requires more labor per unit of capital to produce shoes than computers, so that $L_S/K_S > L_C/K_C$.
  - Example: Cobb-Douglas functions
  - $F_S = L^{0.8}K^{0.2}$, and $F_C = L^{0.3}K^{0.7}$

The Heckscher-Ohlin Model, cont'

Labor Intensity: The demand for labor relative to capital is assumed to be higher in shoes than in computers, $L_S/K_S > L_C/K_C$.



The Heckscher-Ohlin Model, cont'

Assumptions of the Heckscher-Ohlin Model

- Assumption 3: Foreign is labor-abundant, by which we mean that the labor–capital ratio in Foreign exceeds that in Home, $L^*/K^* > L/K$. Equivalently, Home is capital-abundant, so that $K/L > K^*/L^*$.

The Heckscher-Ohlin Model, cont'

Assumptions of the Heckscher-Ohlin Model

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- Assumption 4: The final outputs, shoes and computers, can be traded freely (i.e., without any restrictions) between nations, but labor and capital do not move between countries.

The Heckscher-Ohlin Model, cont'

Assumptions of the Heckscher-Ohlin Model

- Assumption 3: Foreign is labor-abundant, by which we mean that the labor–capital ratio in Foreign exceeds that in Home, $L^*/K^* > L/K$. Equivalently, Home is capital-abundant, so that $K/L > K^*/L^*$.
- Assumption 4: The final outputs, shoes and computers, can be traded freely (i.e., without any restrictions) between nations, but labor and capital do not move between countries.
- Assumption 5: The technologies used to produce the two goods are identical across the countries.
- Assumption 6: Consumer tastes are the same across countries, and preferences for computers and shoes do not vary with a country's level of income.

The Heckscher-Ohlin Model, cont'

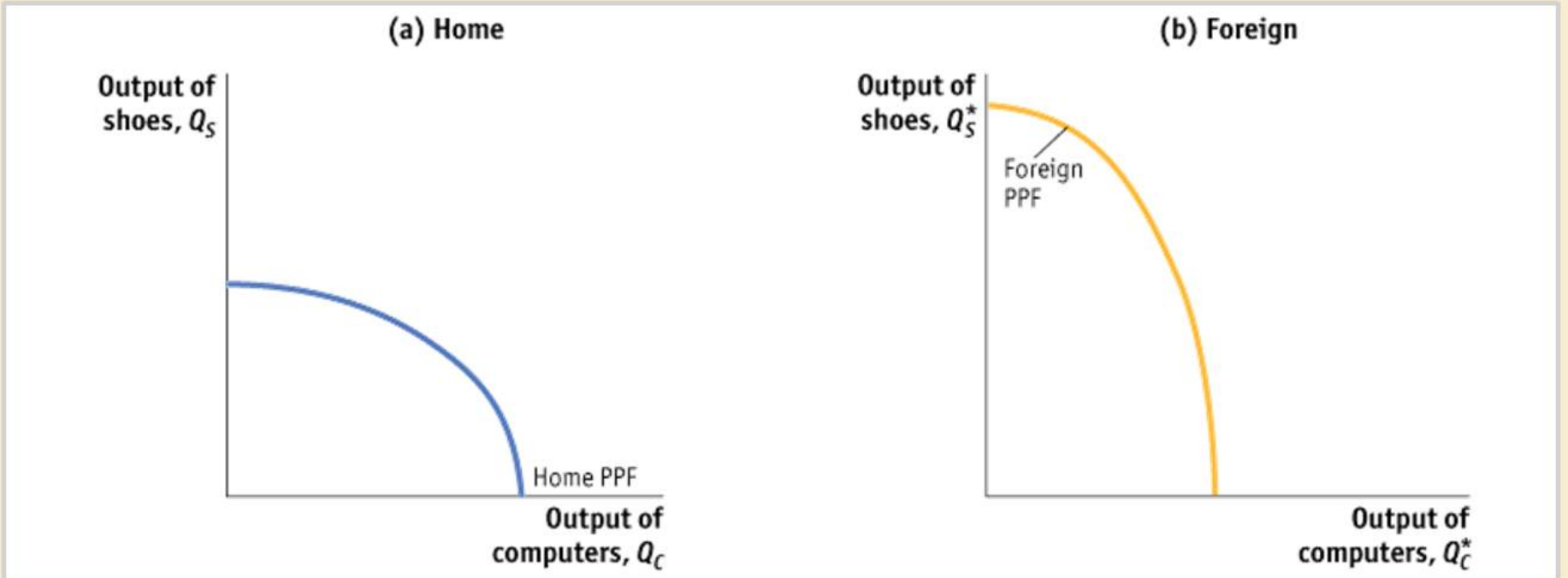
More about Assumptions of the Heckscher-Ohlin Model

- Are factor intensities the same across countries?
- In the United States, agriculture is capital-intensive. In many developing countries, it is labor-intensive.
- This example illustrates a reversal of factor intensities between the two countries.
- In this case, Heckscher-Ohlin is a bit more complicated. We will not consider it in this course.

The Heckscher-Ohlin Model, cont'

No-Trade Equilibrium: PPF, Indifference Curves, Autarky Good Price

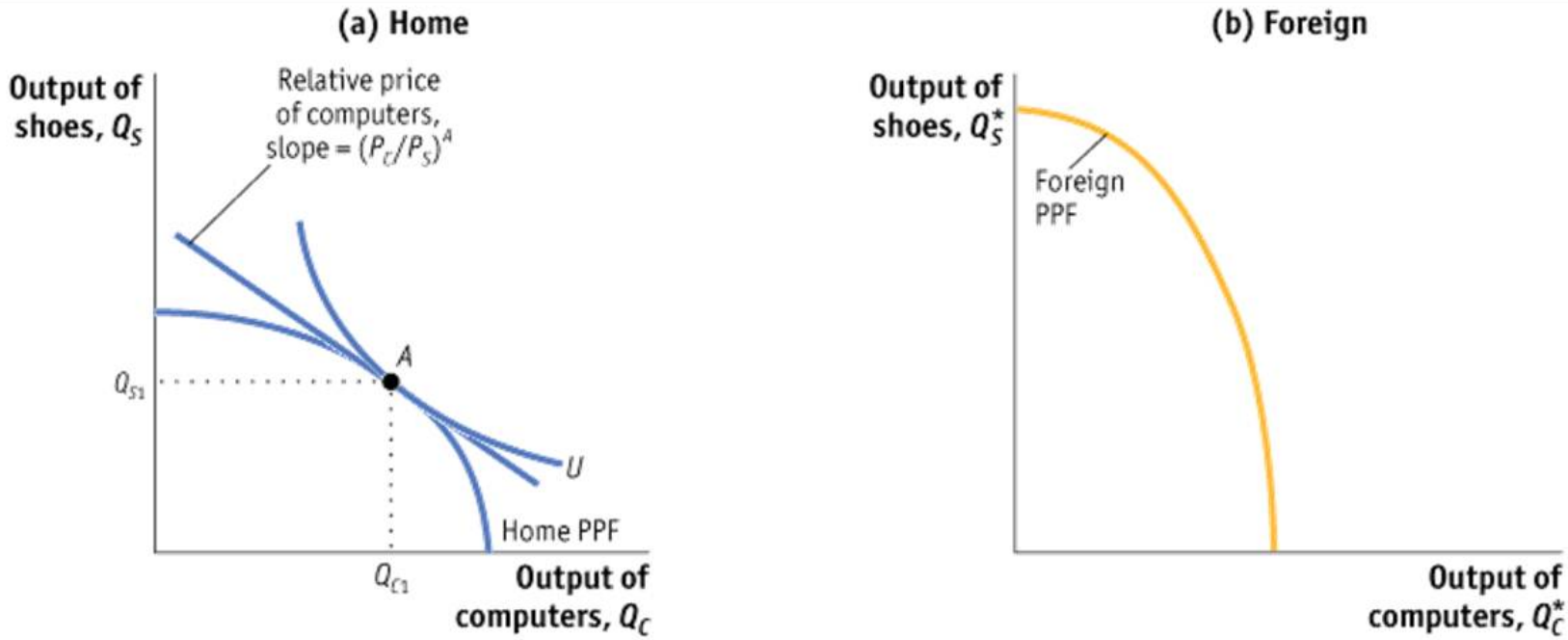
FIGURE 4-2 (1 of 3) **No-Trade Equilibria in Home and Foreign**



Because Home is capital abundant and computers are capital intensive, the Home PPF is skewed toward computers. The reverse is true for Foreign.

The Heckscher-Ohlin Model, cont'

FIGURE 4-2 (2 of 3) **No-Trade Equilibria in Home and Foreign (continued)**

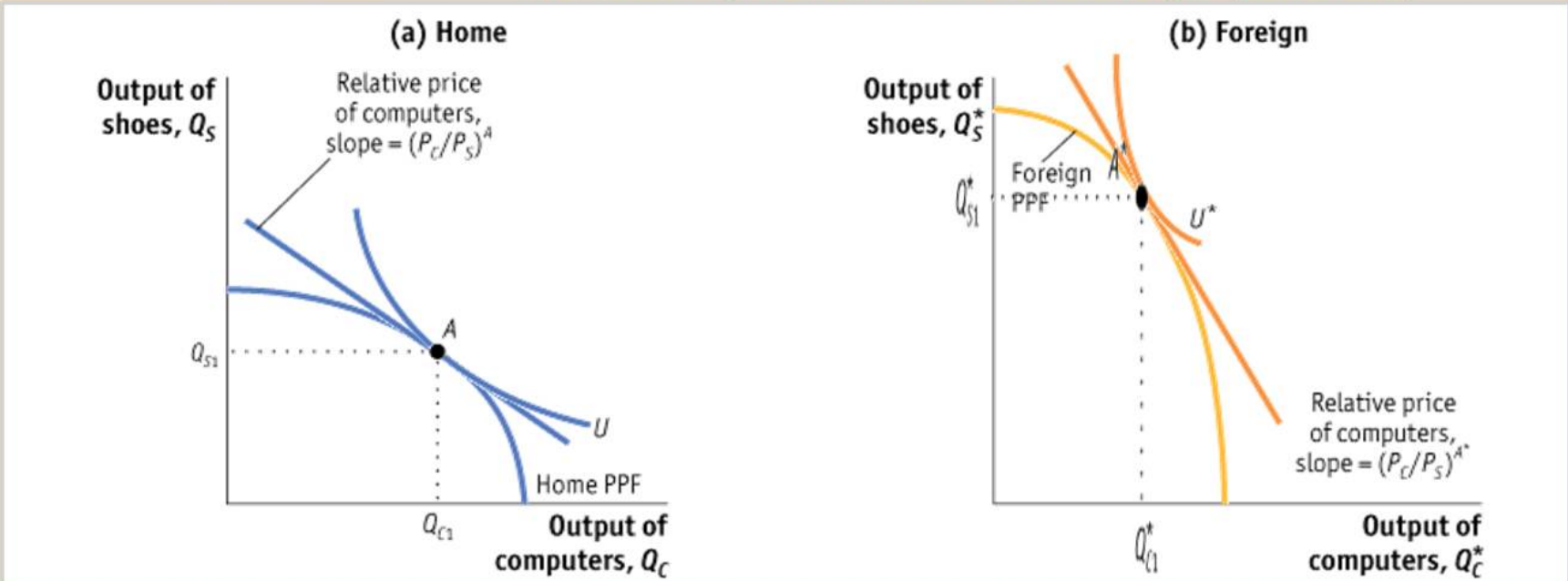


Home preferences are summarized by the indifference curve, *U*. The Home no-trade (or autarky) equilibrium is at point *A*.

The Heckscher-Ohlin Model, cont'

Autarky Equilibrium without Trade:

FIGURE 4-2 (3 of 3) **No-Trade Equilibria in Home and Foreign (continued)**

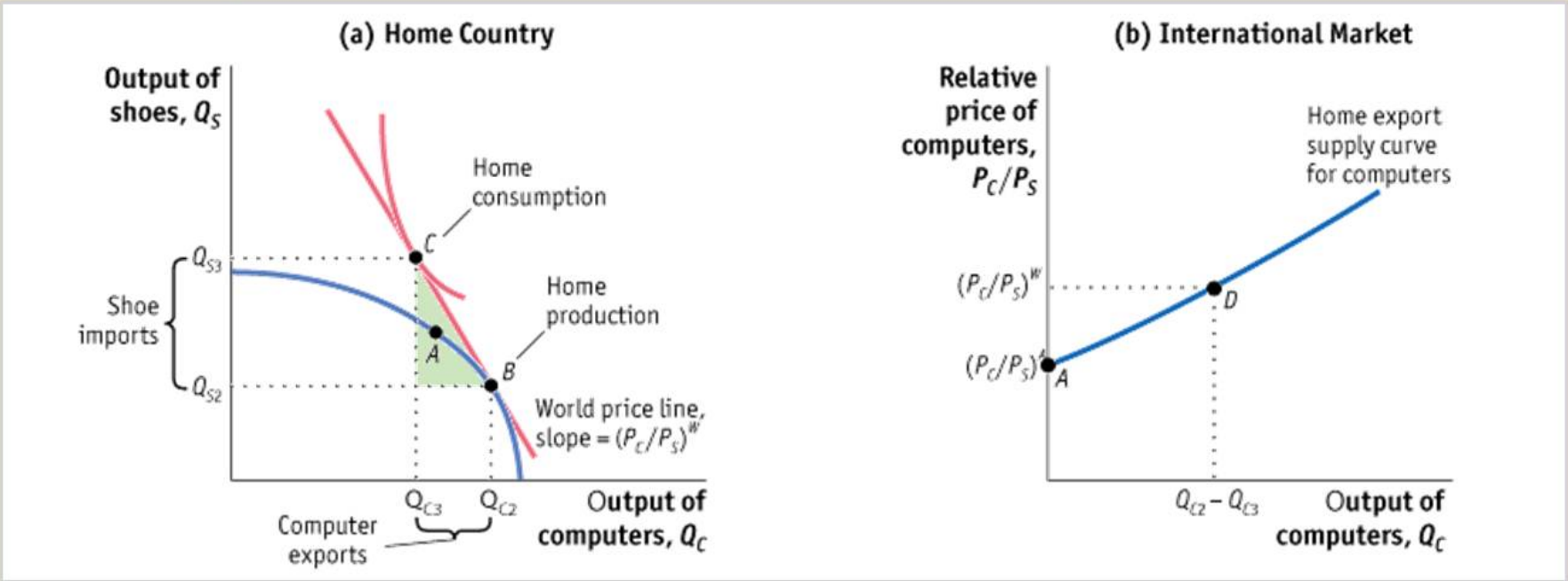


Foreign is labor-abundant and shoes are labor- intensive, so the Foreign PPF is skewed toward shoes. Foreign preferences are summarized by the indifference curve, U^*. The Foreign no-trade equilibrium is at point A^*, with a higher relative price of computers, as indicated by the steeper slope of $(P^*_C/P^*_S)^{A^*}$.

The Heckscher-Ohlin Model, cont'

Home Equilibrium with Free Trade:

FIGURE 4-3 International Free-Trade Equilibrium at Home



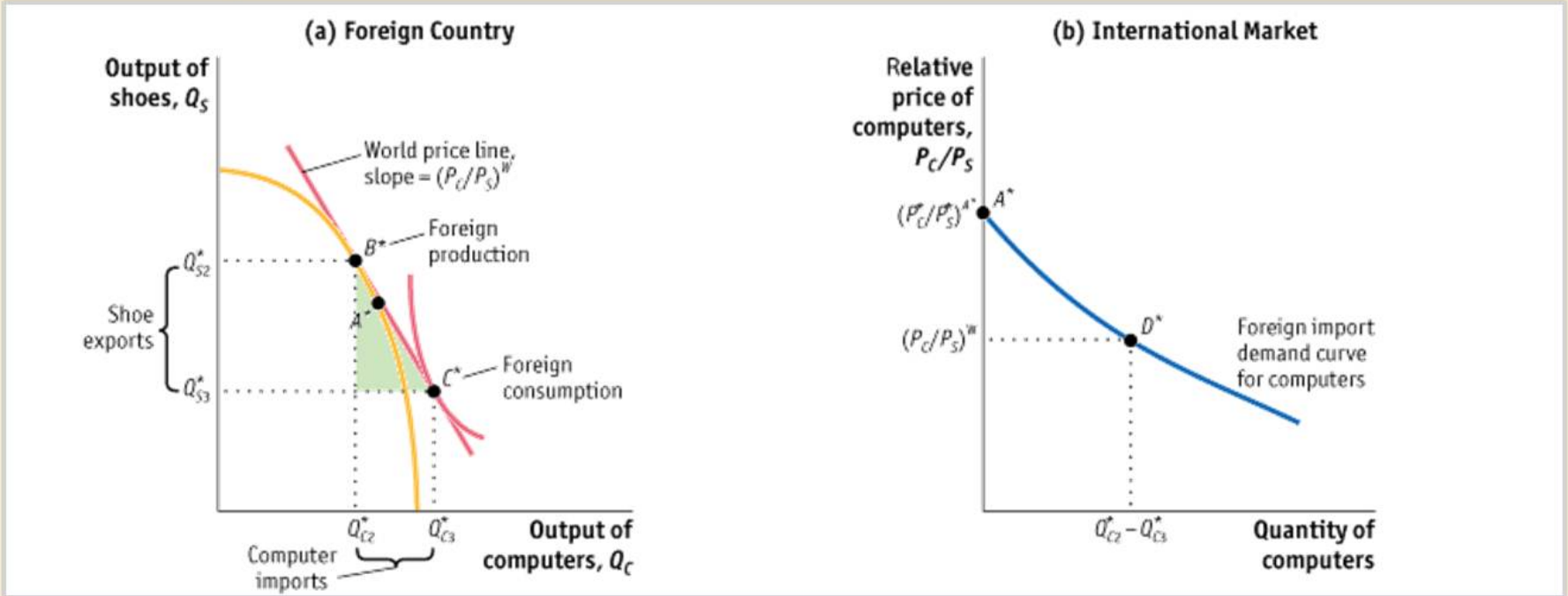
The height of this triangle is the Home imports of shoes (the difference between the amount consumed of shoes and the amount produced with trade, $Q_{S3} - Q_{S2}$).

In panel (b), we show Home exports of computers equal to zero at the no-trade relative price, $(P_C/P_S)^A$, and equal to $(Q_{C2} - Q_{C3})$ at the free-trade relative price, $(P_C/P_S)^W$.

The Heckscher-Ohlin Model, cont'

Foreign Equilibrium with Free Trade:

FIGURE 4-4 International Free-Trade Equilibrium in Foreign



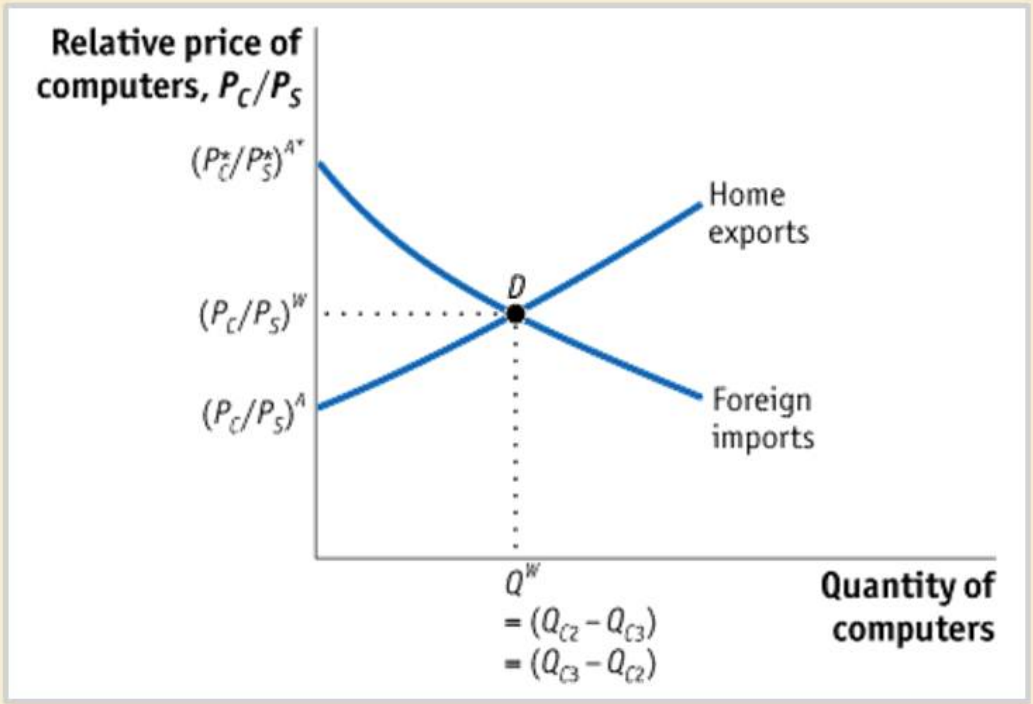
The height of this triangle is Foreign exports of shoes (the difference between the production of shoes and the amount consumed with trade, $Q^*_{S2} - Q^*_{S3}$).

In panel (b), we show Foreign imports of computers equal to zero at the no-trade relative price, $(P^*_C/P^*_S)^{A*}$, and equal to $(Q^*_{C3} - Q^*_{C2})$ at the free-trade relative price, $(P_C/P_S)^W$.

The Heckscher-Ohlin Model, cont'

Equilibrium Price with Free Trade: exports equal imports, there is no reason for the relative price to change and so this is a free-trade equilibrium.

FIGURE 4-5 Determination of the Free-Trade World Equilibrium Price



The world relative price of computers in the free-trade equilibrium is determined at the intersection of the Home export supply and Foreign import demand, at point *D*.

At this relative price, the quantity of computers that Home wants to export, $(Q_{C2} - Q_{C3})$, equals the quantity of computers that Foreign wants to import, $(Q^*_{C3} - Q^*_{C2})$.

The Heckscher-Ohlin Model, cont'

Free Trade Equilibrium: Pattern of Trade

- Home exports computers, the good that uses intensively the factor of production (capital) found in abundance at Home.
- Foreign exports shoes, the good that uses intensively the factor of production (labor) found in abundance there.
- This important result is called the **Heckscher-Ohlin theorem**.
- The theorem says, under the previous assumptions, countries export goods that use their abundant factor intensively.

Testing the Heckscher-Ohlin Theorem

- The first test of the Heckscher-Ohlin theorem: Wassily Leontief in 1953.
- In 1947 the United States was abundant in capital relative to the rest of the world.
- Thus, from the Heckscher-Ohlin theorem, Leontief expected that the United States would export capital-intensive goods and import labor-intensive goods.
- What Leontief actually found, however, was just the opposite: the capital–labor ratio for U.S. imports was higher than the capital–labor ratio found for U.S. exports!
- Leontief's paradox

Testing the Heckscher-Ohlin Theorem, cont'

Leontief's Paradox

TABLE 4-1 **Leontief's Test**

Leontief used the numbers in this table to test the Heckscher-Ohlin theorem. Each column shows the amount of capital or labor needed to produce $1 million worth of exports from, or imports into, the United States in 1947. As shown in the last row, the capital–labor ratio for exports was less than the capital–labor ratio for imports, which is a paradoxical finding.

	Exports	Imports
Capital ($ millions)	$2.55	$3.1
Labor (person-years)	182	170
Capital/labor ($/person)	$14,000	$18,200

Testing the Heckscher-Ohlin Theorem, cont'

Leontief's Paradox: Explanations

- U.S. and foreign technologies are not the same, in contrast to what the HO theorem and Leontief assumed.
- By focusing only on labor and capital, Leontief ignored land abundance in the United States.
- Leontief should have distinguished between skilled and unskilled labor (because it would not be surprising to find that U.S. exports are intensive in skilled labor).
- The data for 1947 may be unusual because World War II had ended just two years earlier.
- The United States was not engaged in completely free trade, as the Heckscher-Ohlin theorem assumes.
- Other explanations...

Testing the Heckscher-Ohlin Theorem, cont'

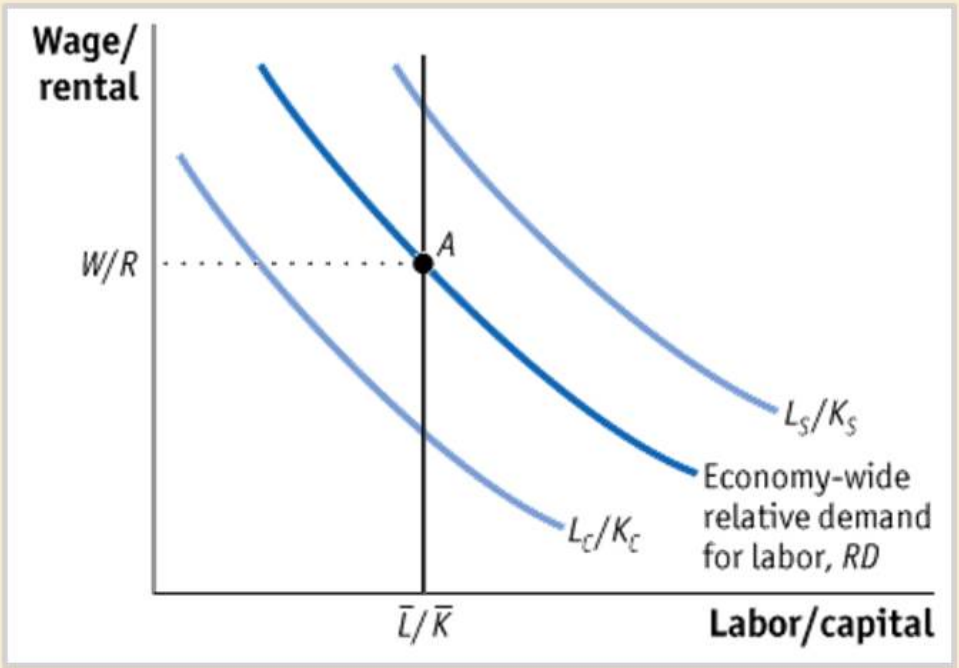
Factor Endowment and Effective Factor Endowment:

- If a country's share of a factor exceeds its share of world GDP, then we conclude that the country is **abundant in that factor**, and if its share in a certain factor is less than its share of world GDP, then we conclude that the country is **scarce in that factor**.
- To allow factors of production to differ in their productivities across countries, we define the effective factor endowment as the actual amount of a factor found in a country times its productivity.
- If a country's share of an effective factor exceeds its share of world GDP, then we conclude that the country is **abundant in that effective factor**; if its share of an effective factor is less than its share of world GDP, then we conclude that the country is **scarce in that effective factor**.

Effects of Trade on Factor Prices

Economy-Wide Relative Demand for Labor at Home

FIGURE 4-10 Determination of Home Wage/Rental



$$\underbrace{\frac{\bar{L}}{\bar{K}}}_{\text{Relative supply}} = \frac{L_C + L_S}{\bar{K}} = \underbrace{\frac{L_C}{K_C}\cdot\left(\frac{K_C}{\bar{K}}\right) + \frac{L_S}{K_S}\left(\frac{K_S}{\bar{K}}\right)}_{\text{Relative demand}}$$

The economy-wide relative demand for labor, *RD*, is an average of the L_C/K_C and L_S/K_S curves and lies between these curves.

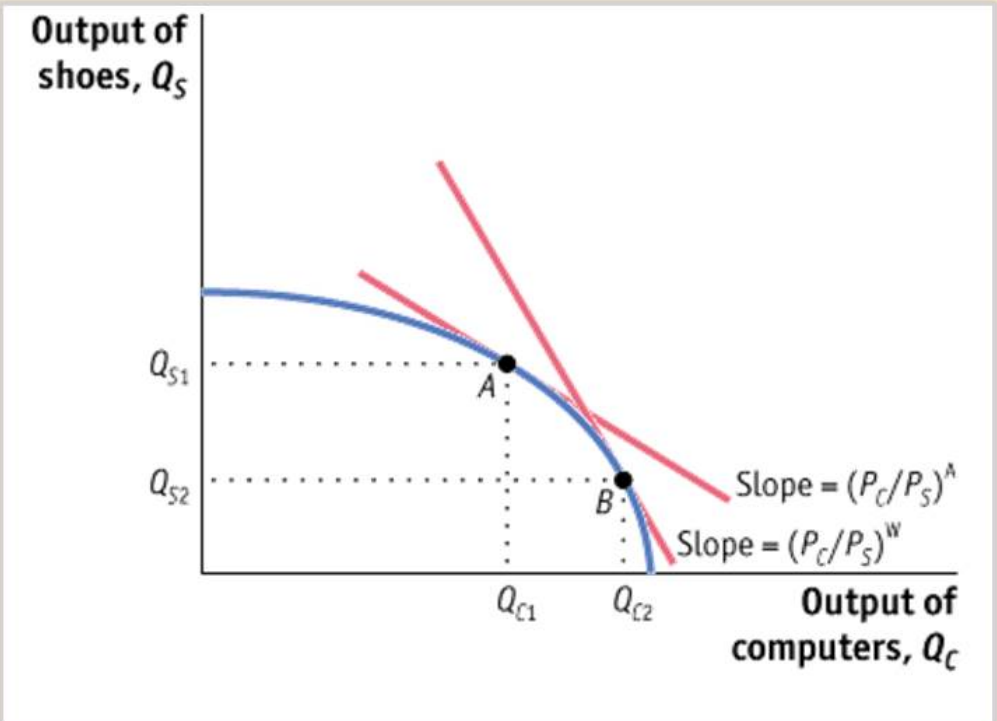
The relative supply, *L/K*, is shown by a vertical line because the total amount of resources in Home is fixed.

The equilibrium point *A*, at which relative demand *RD* intersects relative supply *L/K*, determines the wage relative to the rental, *W/R*.

Effects of Trade on Factor Prices

Home Production with Trade

FIGURE 4-11 Increase in the Price of Computers in Trade



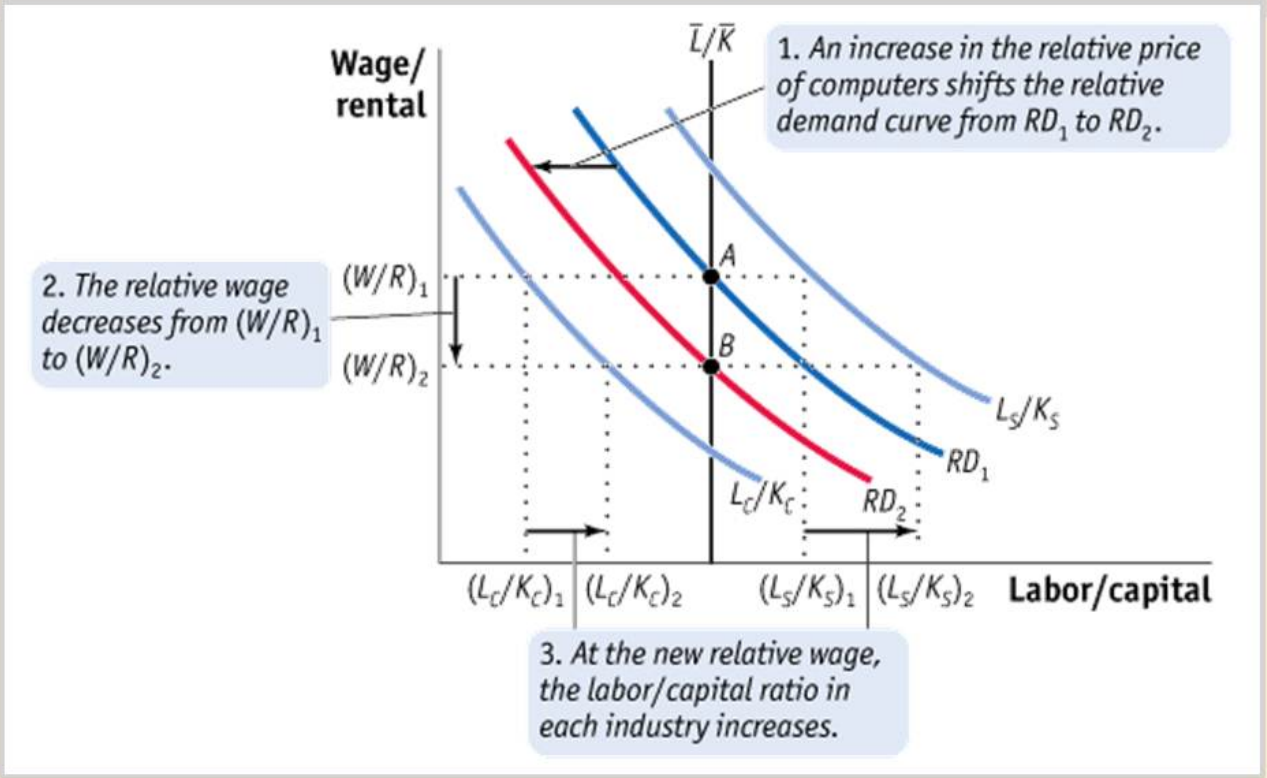
An increase in the relative price of computers to the world price, as illustrated by the steeper world price line, $(P_C/P_S)^W$, shifts production from point *A* to *B*.

At point *B*, there is a higher output of computers and a lower output of shoes, $Q_{C2} > Q_{C1}$ and $Q_{S2} < Q_{S1}$.

Effects of Trade on Factor Prices

Relative Demand for Labor at Home in Trade

FIGURE 4-12 Effect of a Higher Relative Price of Computers on Wage/Rental



As a result, the wage relative to the rental falls from $(W/R)_1$ to $(W/R)_2$.

The lower relative wage causes both industries to increase their labor–capital ratios, as illustrated by the increase in both L_C/K_C and L_S/K_S at the new relative wage.

$$\frac{\bar{L}}{\bar{K}} = \frac{L_C}{K_C}\cdot\left(\frac{K_C}{\bar{K}}\right) + \frac{L_S}{K_S}\left(\frac{K_S}{\bar{K}}\right)$$

$\uparrow$ $\uparrow$ $\uparrow$ $\downarrow$

Relative supply: No change

Relative demand: No change in total

Effects of Trade on Factor Prices, cont'

Determination of the Real Wage and Real Rental:

- Real Rental
  - $R = P_C \cdot MPK_C$, $R = P_S \cdot MPK_S$
  - $MPK_C = R/P_C \uparrow$, $MPK_S = R/P_S \uparrow$
- Real Wage
  - $W = P_C \cdot MPL_C$, $W = P_S \cdot MPL_S$
  - $MPL_C = W/P_C \downarrow$, $MPL_S = W/P_S \downarrow$

Effects of Trade on Factor Prices, cont'

Determination of the Real Wage and Real Rental:

- **Stolper-Samuelson Theorem:** In the long run, when all factors are mobile, an increase in the relative price of a good will increase the real earnings of the factor used intensively in the production of that good and decrease the real earnings of the other factor.

- For our example, the **Stolper-Samuelson theorem** predicts that when Home opens to trade and faces a higher relative price of computers, the real rental on capital in Home rises and the real wage in Home falls. In Foreign, the changes in real factor prices are just the reverse.

Summary

Key Points:

1. In the Heckscher-Ohlin model, we assume that the technologies are the same across countries and that countries trade because the available resources (labor, capital, and land) differ across countries.

2. The Heckscher-Ohlin model is a long-run framework, so labor, capital, and other resources can move freely between the industries.

3. With two goods, two factors, and two countries, the Heckscher-Ohlin model predicts that a country will export the good that uses its abundant factor intensively and import the other good.

Summary, cont'

4. The first test of the Heckscher-Ohlin model was made by Leontief using U.S. data for 1947. He found that U.S. exports were less capital-intensive and more labor-intensive than U.S. imports. This was a paradoxical finding because the United States was abundant in capital.

5. The assumption of identical technologies used in the Heckscher-Ohlin model does not hold in practice. Current research has extended the empirical tests of the Heckscher-Ohlin model to allow for many factors and countries, along with differing productivities of factors across countries. When we allow for different productivities of labor in 1947, we find that the United States is abundant in effective—or skilled—labor, which explains the Leontief paradox.

Summary, cont'

6. According to the Stolper-Samuelson theorem, an increase in the relative price of a good will cause the real earnings of labor and capital to move in opposite directions: the factor used intensively in the industry whose relative price goes up will find its earnings increased, and the real earnings of the other factor will fall.

7. Putting together the Heckscher-Ohlin theorem and the Stolper-Samuelson theorem, we conclude that a country's abundant factor gains from the opening of trade (because the relative price of exports goes up), and its scarce factor loses from the opening of trade.